Exercises

1. Image creation

1. Create an image of 100x100 pixels, consisting on a chessboard of 10x10, with cells black and white.
2. Now make the cells have several different colors... white, black, green, red, blue, yellow, orange, etc.
3. Create an image of 100x100 pixels with white background and one blue circle of radius 30 pixels centered in the center of the image.
4. **Perhaps part of PC1**: Create an image of the fractal of Mandelbrot (see details below)

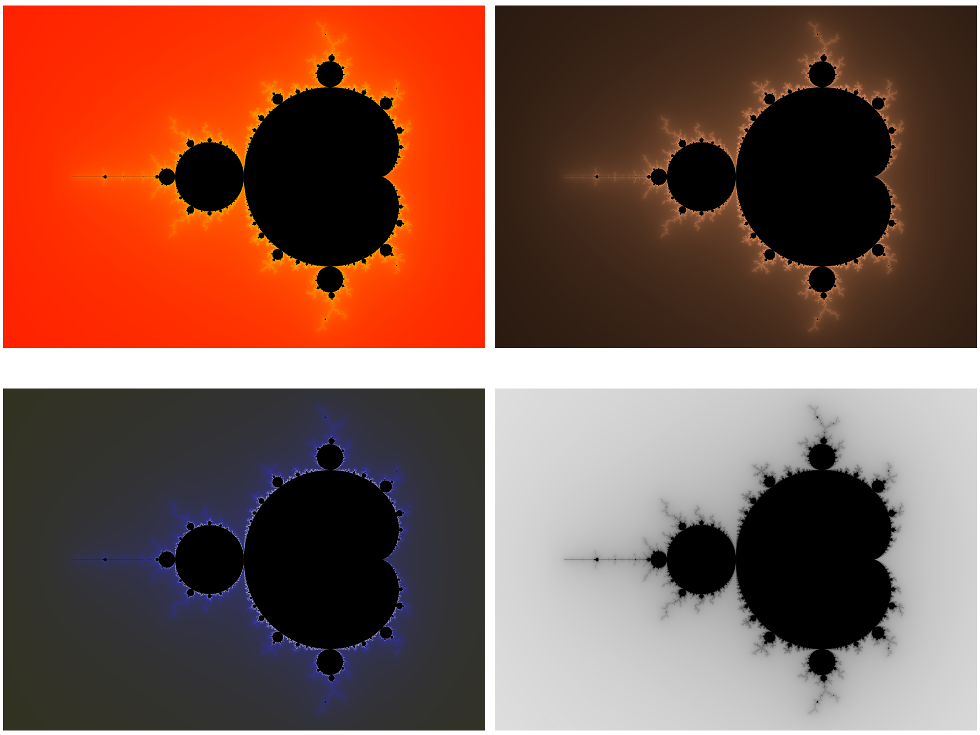
2. Simple animations

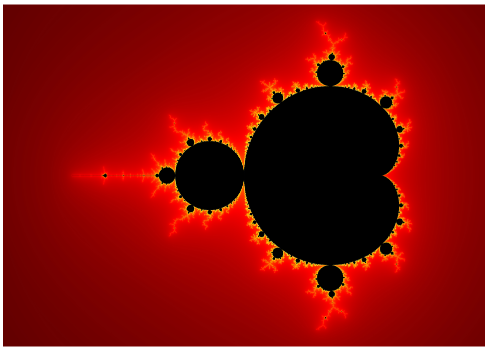
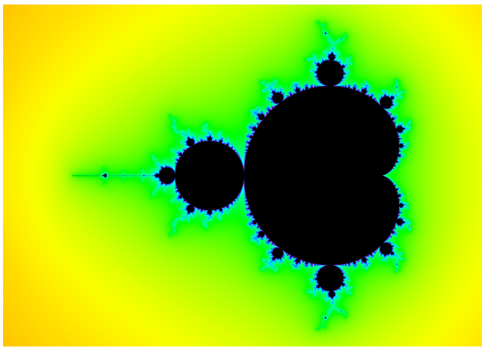
1. Use 1.3 to generate a simple animation in which the blue circle moves around and when it hits one border of the image, it *bounces*.
2. **Perhaps part of PC1**: Create an animation showing the *game of life* in a board of size $N \times M$.

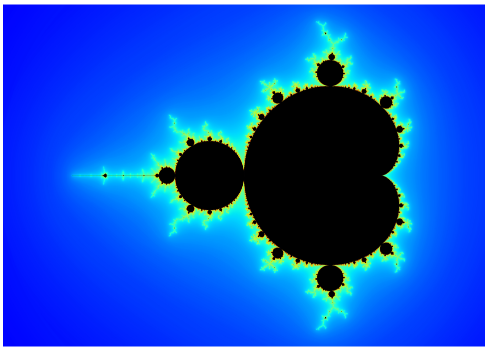
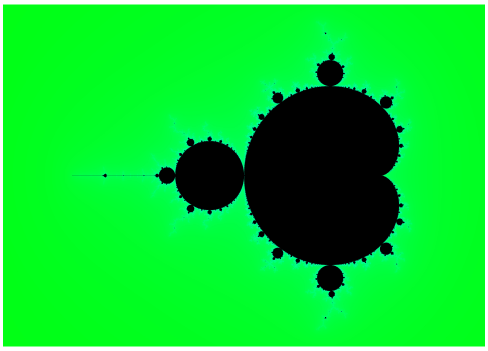
On the Mandelbrot fractal of 1.d

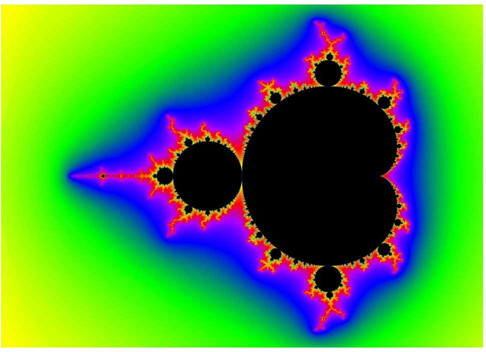
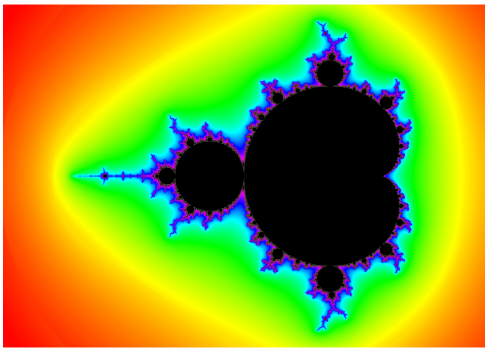
Please refer to https://en.wikipedia.org/wiki/Mandelbrot_set for the explanation of the algorithm to colorize each pixel in the image

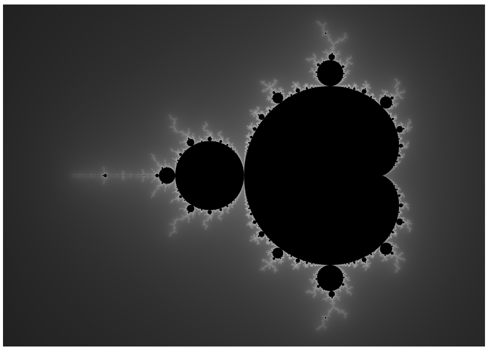
Examples





On the *Game of Life* of 2.d

Please refer to https://en.wikipedia.org/wiki/Conway%27s_Game_of_Life for the explanation on what is expected